Week 7 findings memo

30 October 2026 | Exploratory E3 | Scientific freeze 6c97d939

Completion

The fixed 53-path diagnostic selection reproduced all 159 original estimator records. The independent follow-up completed 20,000 paths / 60,000 fits; the floor sensitivity completed 24,000 physical realizations / 72,000 fits, representing 8,000 independent coupling groups. Both new studies passed full audits, actual figure review and completion seals, with zero pipeline errors. E1/E2 files and decisions remain unchanged. [1]

Main findings

Boundaries and curvature. All 85 original point-invalid records are consistent with the saved conditional-regression boundary evidence. All 14 evaluated exact profiles match their stored fits within the fixed diagnostic resolution. The rejected irregular exact interval reproduces its whitened Hessian discrepancy in longdouble: 0.095654 versus the unchanged 0.01 gate. Approximately fourfold decay on step halving supports finite-difference truncation amplified by weak curvature. No original interval is replaced; finite searches do not prove global optimality. [2]

Independent follow-up, n=250 and q=2. The study retains 421 invalid points and 23 further unavailable intervals: 444 records on 231 source paths. One irregular exact candidate remains boundary_suspected; no finite optimum is certified. Operational theta coverage is 92.63% for regular exact/PFML, 94.77% for irregular exact, 72.29% for irregular PFML and 0% observed for Euler. Regular exact/PFML conditional coverage is 94.64%; excluding unavailable intervals would hide that denominator difference. Irregular exact theta's Wilson interval is 94.32%-95.19%; irregular PFML's is 71.40%-73.16%. Euler's zero count has a 0.0384% Wilson upper limit. New results do not overwrite headline classifications. [3]

Floor sensitivity, n=1000 and q=0.5. All 72,000 points and intervals are valid. Across the fixed CV/epsilon contrasts, the largest estimated RMSE-ratio change is exact theta at CV=3 and epsilon=1e-4: 1.00023261, with paired 95% Monte Carlo interval [0.99974475, 1.00072070]. The largest absolute observed coverage difference is 0.05 percentage points. Small nonzero bias/RMSE effects are retained. These results support modest sensitivity on this grid, not universal invariance or formal equivalence. CV=0 is one shared exact control; other variants share innovations, not one continuous trajectory. [4]

Assurance and limits

Independent saved-data review reproduced all 126 parameter-summary rows and 72 epsilon contrasts; a separate review passed 4,396 checks on selected diagnostics, including algebra on 352 saved Hessians. All eight PNGs were visually inspected. Current archive hashes verify, but the earlier unattributed manifest-state incidents remain a historical operational limitation. Nonrecurrence does not establish their cause. Week 8 requires a separate committed gate after Week 7 publication. No new core estimator or inference defect was established by these checks. [1,2]

Evidence

1. Completion report and verification.
2. Selected numerical disposition and exact review scope.
3. Follow-up summaries and retained outcomes.
4. Paired floor contrasts and source audit.